

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 40%

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

11. Detection Model Comparison under Resource Constraints

Performance:

Model	Accuracy	Latency
A	85%	40 ms
B	92%	90 ms
C	90%	60 ms

Compare the models and evaluate which is optimal for real-time systems with moderate accuracy requirements. Justify your decision.

ANSWER:

1. Understanding of Data

The three models show a clear **trade-off between accuracy and latency**:

- **Model A:** Lowest accuracy (85%) but fastest response time (40 ms).
- **Model B:** Highest accuracy (92%) but highest latency (90 ms).

- **Model C:** Accuracy of 90% with moderate latency of 60 ms.

Thus, increasing accuracy generally results in increased processing time.

2. Comparative Analysis

Model A:

Model A is computationally efficient because it has the lowest latency of **40 ms**. It is suitable when response speed is the highest priority. However, its 85% accuracy is the lowest among the three models, which may lead to more incorrect detections.

Model B:

Model B provides the highest accuracy of **92%**, making it the most reliable in terms of detection performance. However, its **90 ms latency** is more than twice that of Model A. This makes it less desirable for applications requiring fast real-time responses.

Model C:

Model C provides **90% accuracy with 60 ms latency**. It sacrifices only 2 percentage points of accuracy compared with Model B while reducing latency by **30 ms**. It is therefore a better balance between accuracy and computational efficiency.

3. Application Context

For a **real-time system with moderate accuracy requirements**, latency is an important factor because the system must respond quickly. At the same time, accuracy should be sufficiently high to ensure reliable detection.

Model A is very fast but has relatively lower accuracy. Model B is highly accurate but comparatively slow. **Model C provides the best compromise**, offering high enough accuracy while maintaining reasonably low latency.

4. Decision

Model C is the optimal choice.

Factor	Model A	Model B	Model C
Accuracy	85%	92%	90%
Latency	40 ms	90 ms	60 ms
Computational efficiency	Excellent	Low	Good
Suitability for real-time use	Good	Moderate	Very Good
Overall choice	2nd	3rd	1st

Conclusion

Model C should be selected for the real-time system. It achieves **90% accuracy with 60 ms latency**, providing an effective balance between detection performance and computational requirements. Although Model A is faster and Model B is more accurate, Model C avoids the major weakness of each: it provides better accuracy than A without the high latency of B.

Therefore, considering **accuracy, robustness, response time, and resource constraints**, **Model C is the most suitable model for a real-time system with moderate accuracy requirements.**

12. Robustness vs Speed Trade-off

Performance:

Method	Accuracy	Robustness	Speed
A	88%	High	Slow
B	85%	Moderate	Fast
C	90%	Low	Moderate

Compare the methods and evaluate the best choice for a balanced system. Justify trade-offs.

ANSWER :

1. Understanding of Data

The three methods demonstrate a trade-off between accuracy, robustness, and speed:

- Method A has good accuracy (88%) and high robustness, but it is slow.
- Method B is the fastest, but has the lowest accuracy (85%) and only moderate robustness.
- Method C provides the highest accuracy (90%) and moderate speed, but its robustness is low.

2. Comparative Analysis

MethodA:

Method A is highly robust, meaning it can maintain reliable performance under changing or difficult conditions. Its 88% accuracy is also good. However, its main disadvantage is slow processing speed, which can affect real-time applications.

MethodB:

Method B has the highest speed, making it suitable for applications where quick response is critical. However, its accuracy is only 85%, and its robustness is moderate. Therefore, it may not provide sufficiently reliable results in varying operating conditions.

MethodC:

Method C has the highest accuracy at 90% and moderate speed. However, its low robustness is a major limitation. Its performance may degrade when input conditions change, such as noise, illumination, or environmental variations.

3. Trade-off Analysis

For a **balanced system**, all three factors should be considered:

Factor	A	B	C
Accuracy	Good	Lowest	Highest
Robustness	High	Moderate	Low
Speed	Slow	Fastest	Moderate
Overall balance	Best	Speed-focused	Accuracy-focused

Method C should not automatically be selected just because it has the highest accuracy. Its low robustness can make it unreliable in real-world conditions.

4. Best Choice

Method A is the best choice for a balanced system, especially when reliability under changing conditions is important.

It provides **88% accuracy and high robustness**, which is a strong combination. Although it is slower than B and C, the loss in speed can be acceptable when the application does not require extremely fast real-time processing.

If the application is **strictly real-time**, Method B could be preferred because of its fast speed. If **maximum accuracy** is the priority and operating conditions are controlled, Method C may be selected.

Conclusion

Method A is the most balanced overall choice because it combines good accuracy with **high robustness**. Its slow speed is the main trade-off, but the increased reliability can be more valuable than a small improvement in processing speed. Therefore, for applications where **consistent and reliable performance is more important than maximum speed**, Method A is recommended

13. Optical Flow Stability Analysis

Results:

Method	Accuracy	Noise Sensitivity
A	High	High
B	Moderate	Low

Compare both methods and evaluate suitability for real-time tracking in noisy environments.

ANSWER:

Comparison

Method A provides high accuracy under normal conditions, but its high noise sensitivity means that its performance can deteriorate significantly when the video contains noise.

Method B has moderate accuracy but low noise sensitivity. Therefore, it can maintain more stable performance when the input contains noise.

Evaluation

For **real-time tracking in noisy environments**, **Method B is the better choice**. Although it sacrifices some accuracy, its low sensitivity to noise provides greater stability and reliability. Method A may be preferred in clean environments where maximum accuracy is more important.

Conclusion

Method B is recommended for noisy real-time tracking because robustness to noise is more important than achieving maximum accuracy when environmental conditions are unpredictable.

14. Motion Estimation Accuracy vs Efficiency

Results:

Method	Error	Computation
A	4%	High
B	9%	Low
C	6%	Moderate

Compare methods and evaluate the best approach for embedded systems.

ANSWER:

Comparison

- Method A: Has the lowest error (4%) and therefore the highest estimation accuracy. However, it requires high computational resources.
- Method B: Has the highest error (9%) but requires the least computation, making it highly efficient.
- Method C: Has a moderate error of 6% with moderate computational requirements.

Evaluation

For embedded systems, computational resources are usually limited. Therefore, Method A may be too computationally expensive despite its excellent accuracy.

Method B is computationally efficient but has relatively high error. Method C provides the best balance, reducing the error compared with B while avoiding the high computational requirement of A.

Conclusion

Method C is the best approach for embedded systems because it provides 6% error with moderate computation. It offers a practical compromise between accuracy and computational efficiency.

16. Object Detection under Scale Variation

Results:

Object Size	Accuracy
Large	94%
Medium	80%

Small 55%

Compare detection performance and evaluate system limitations. Justify improvements.

ANSWER:

Comparison

The results show that detection accuracy decreases significantly as object size decreases.

- Large objects: 94% accuracy, indicating reliable detection.
- Medium objects: 80% accuracy, showing a noticeable reduction.
- Small objects: Only 55% accuracy, indicating poor detection performance.

The difference between large and small objects is 39 percentage points, demonstrating a significant scale-related limitation.

System Limitations

The main limitation is the difficulty of extracting sufficient features from small objects. Small objects occupy fewer pixels, so important details may be lost during image resizing, feature extraction, or downsampling.

Improvements

The system can be improved by:

1. Using **multi-scale feature extraction**.
2. Increasing input image resolution.
3. Using **feature pyramid networks (FPN)** or similar multi-scale architectures.
4. Applying data augmentation with different object sizes.
5. Improving the training dataset with more small-object examples.
6. Using appropriate image preprocessing and super-resolution techniques where necessary.

Conclusion

The system performs well for large objects but has a serious limitation for small objects. Therefore, **improving multi-scale feature representation and training with more small-object samples** would help increase detection accuracy.

20. Surveillance Model Selection

Results:

	Model Accuracy	Latency
A	82%	30 ms
B	90%	70 ms
C	95%	120 ms

Compare and evaluate under strict latency constraints.

ANSWER:

Comparison

- **Model A:** Lowest accuracy but fastest response at 30 ms.
- **Model B:** Provides 90% accuracy with 70 ms latency.
- **Model C:** Highest accuracy at 95%, but has the highest latency of 120 ms.

Evaluation

Under **strict latency constraints**, response time is the primary consideration. Model C may provide excellent accuracy, but its 120 ms latency makes it unsuitable when immediate surveillance responses are required.

Model A is the fastest and is therefore the safest choice when latency is extremely strict. Model B provides considerably better accuracy but requires 40 ms more latency than A.

Conclusion

Model A is the best choice under strict latency constraints because its **30 ms latency** enables fast real-time surveillance. The trade-off is its lower 82% accuracy. If the latency requirement allows approximately 70 ms, **Model B would be a better balanced alternative**, providing 90% accuracy.

21. Noise Impact on Detection

Results:

Noise Level	Accuracy
Low	92%
Medium	80%
High	60%

Compare performance and evaluate system robustness.

ANSWER:

Comparison

The results show a clear negative relationship between noise level and detection accuracy.

- At low noise, accuracy is high at 92%.
- At medium noise, accuracy decreases to 80%.
- At high noise, accuracy falls sharply to 60%.

The accuracy drops by 32 percentage points from low to high noise.

Evaluation

This indicates that the detection system is not highly robust to noise. As noise increases, important image features become distorted or difficult to distinguish, resulting in incorrect detections.

Improvements

System robustness can be improved using:

- Noise filtering and image preprocessing.
- Training with noisy images.
- Data augmentation using different noise levels.
- Robust feature extraction techniques.
- Denoising algorithms before detection.
- Models trained specifically for noisy environments.

Conclusion

The system performs well under low noise but loses significant accuracy under high noise. Therefore, noise-aware training and preprocessing are necessary to improve robustness and maintain reliable detection in real-world conditions.

27. Video Processing Pipeline Comparison

Results:

Pipeline	Accuracy	Speed
A	88%	Fast
B	92%	Moderate
C	95%	Slow

Compare and evaluate suitability.

ANSWER:

Comparison

PipelineA:

It provides the highest processing speed, making it suitable for real-time applications. However, its accuracy is only 88%, which may result in more detection errors.

PipelineB:

It provides **92% accuracy with moderate speed**. It offers a good compromise between processing speed and detection performance.

PipelineC:

It has the highest accuracy of **95%**, but its slow processing speed makes it unsuitable for applications requiring real-time video processing.

Evaluation

The best pipeline depends on the application:

- For **maximum speed**, choose **A**.
- For **maximum accuracy**, choose **C**.
- For a **balanced real-time system**, choose **B**.

Conclusion

Pipeline B is the most suitable overall choice because it achieves **92% accuracy with moderate speed**. It provides a better balance between accuracy and computational efficiency than A and C.

Final recommendation: **B** is preferred when both reliable detection and reasonably fast video processing are required.

Code of Conduct:

*I DARANI E (!92312120) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application ; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure,	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

		visuals, and terminology			
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